

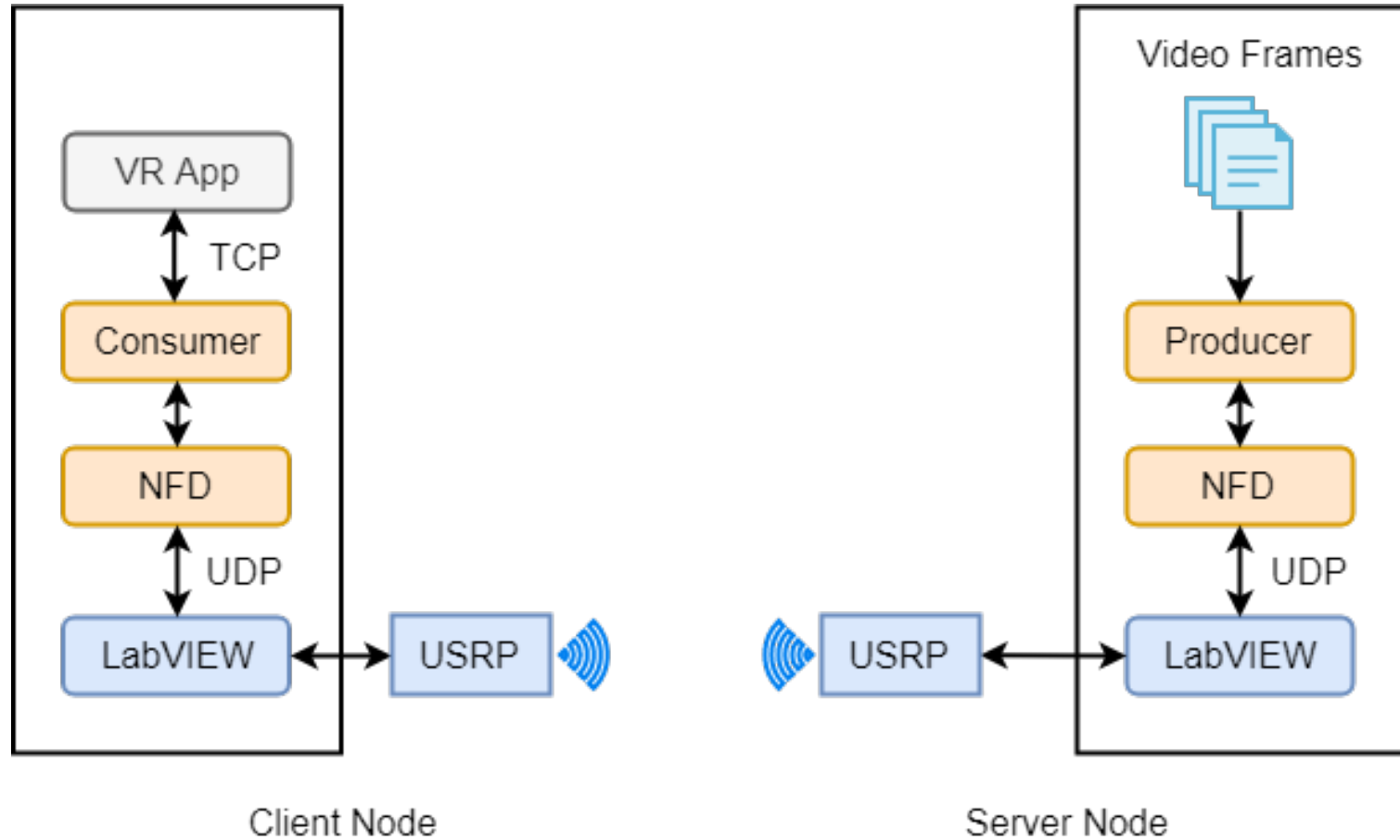
SPLICE: Secure Predictive Low-Latency Information Centric Edge for Next Generation Wireless Networks

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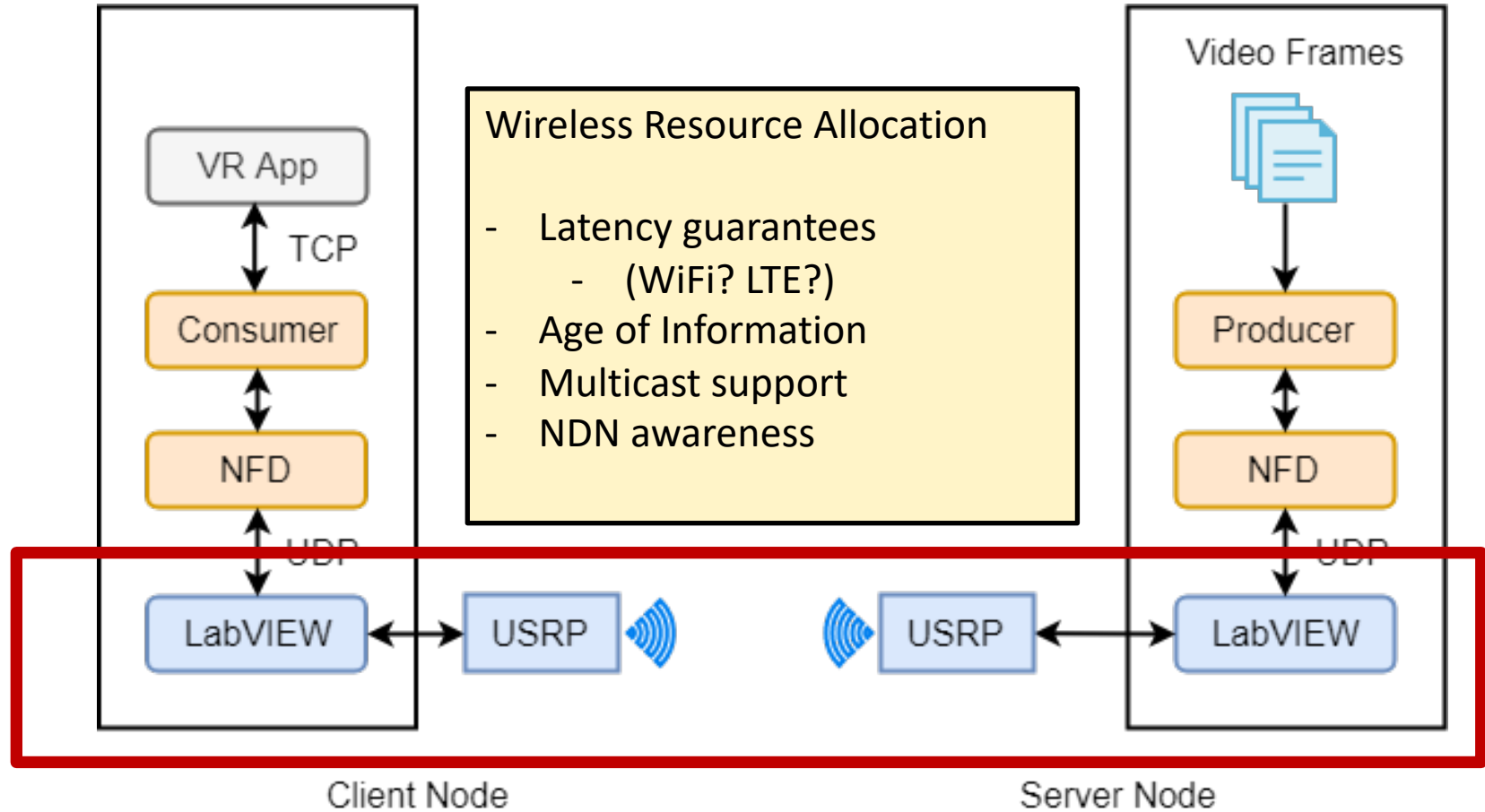
SPLICE Objectives

- Embed ICN support in Wireless Edge Network
 - Exploit commonality/correlation of user information requirements
 - Focusing on extracting desired information
 - Controllable wireless delivery modalities
 - Keep application performance and security in mind

System Architecture



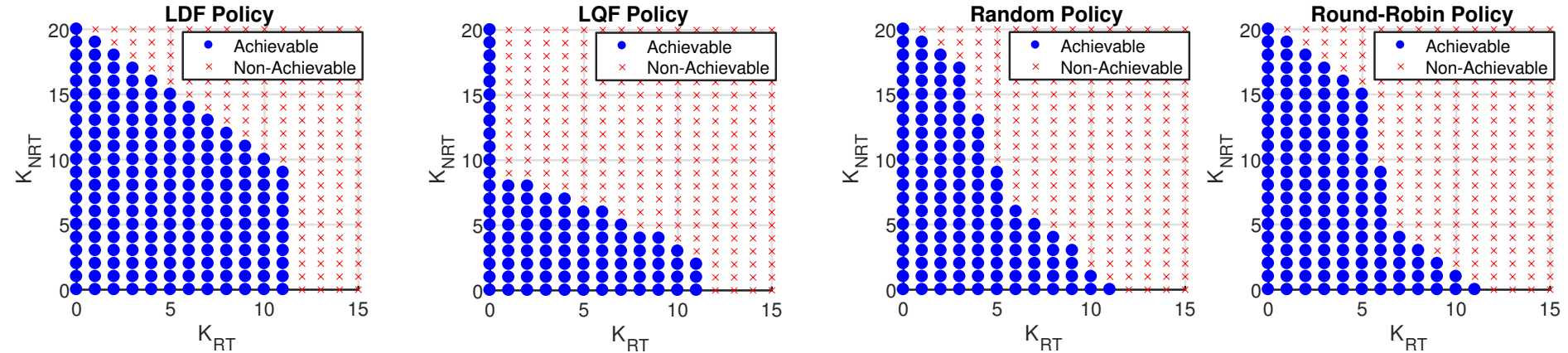
System Architecture



A. Wireless Resource Allocation

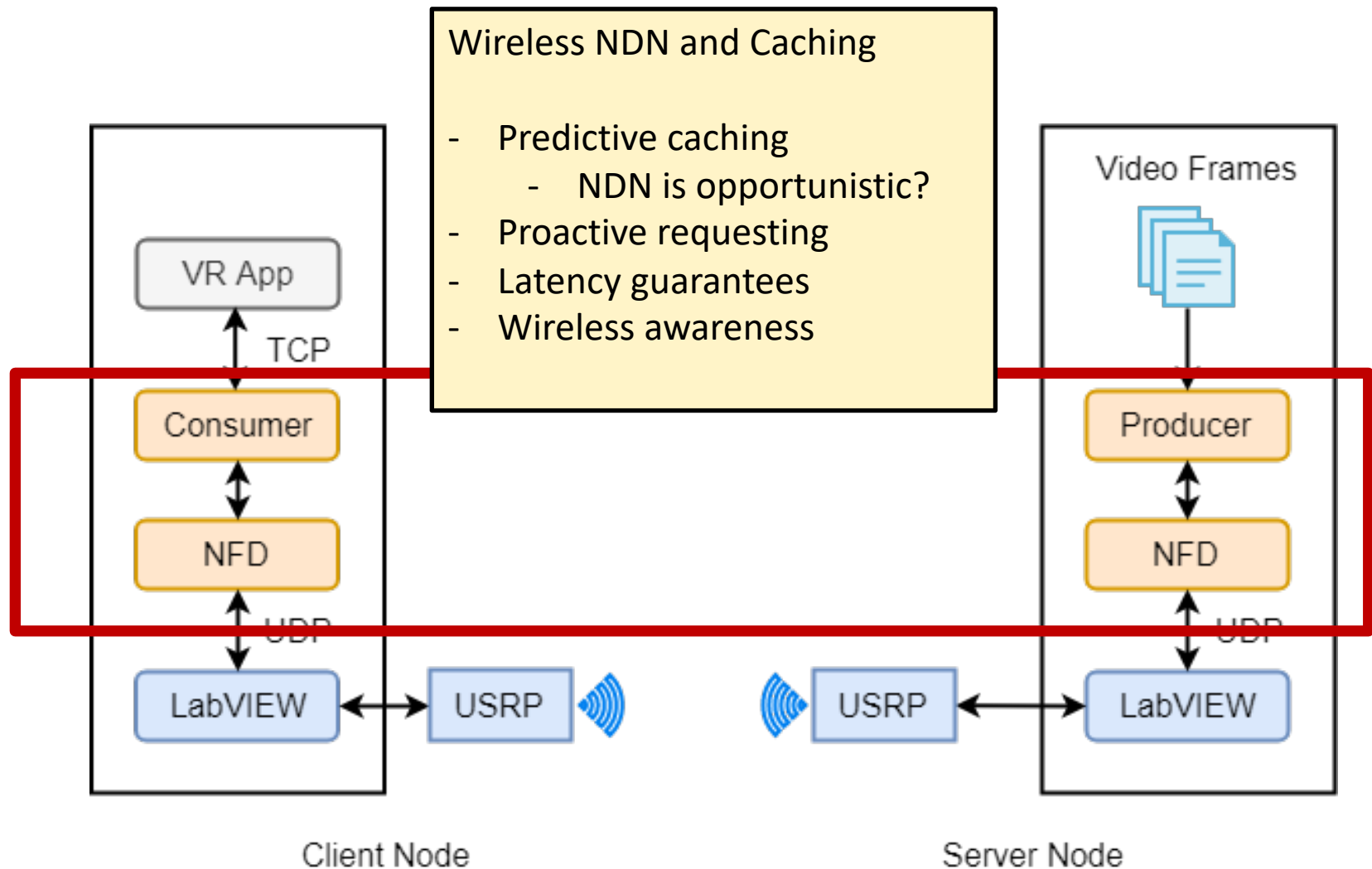
- Delay-based scheduling and empirical studies. Latency < 5 ms, throughput 30-60 Mbps (based on BW)
 - MobiHoc 2018 best paper candidate
- Novel algorithms for broadcast with deadline guarantees
- Uplink scheduling with deadline guarantees
- Credible situational updates by extending Age of Information to multi-packet
- QFlow: Reinforcement Learning for network configuration

PULS: Capacity Regions



- Between 30 – 60 Mbps (based on bandwidth).
- Latencies between 3-5 ms achievable.
- High Spectral efficiency of LDF when compared to RAND (currently used in Wi-Fi).

System Architecture



B. Wireless NDN and Caching

- ICT Notify for proactive requests
- Predictive requests for VR
- NDN forwarding optimization
- Distributed Caching over Unreliable Channels
 - Infocom 2019 Best Paper Award
- Learning rates of caching algorithms
- Dynamic Caching for Multiple Flows
- Proactive Caching for Low Access-Delay Services

Predictive Caching for VR



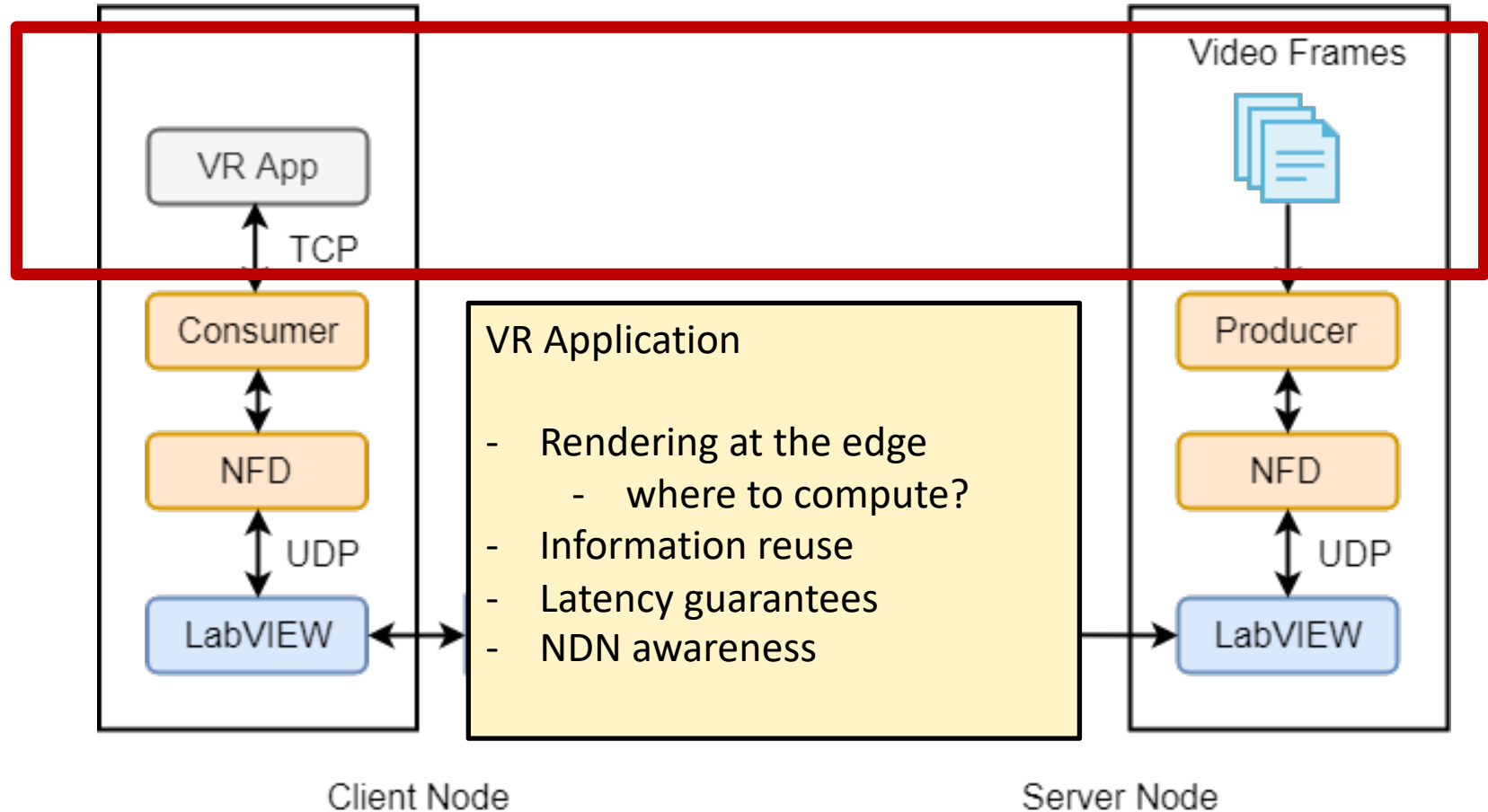
(a) Predictive Scheduling Policy

- Addition of prediction to NDN is able to enable services with higher performance

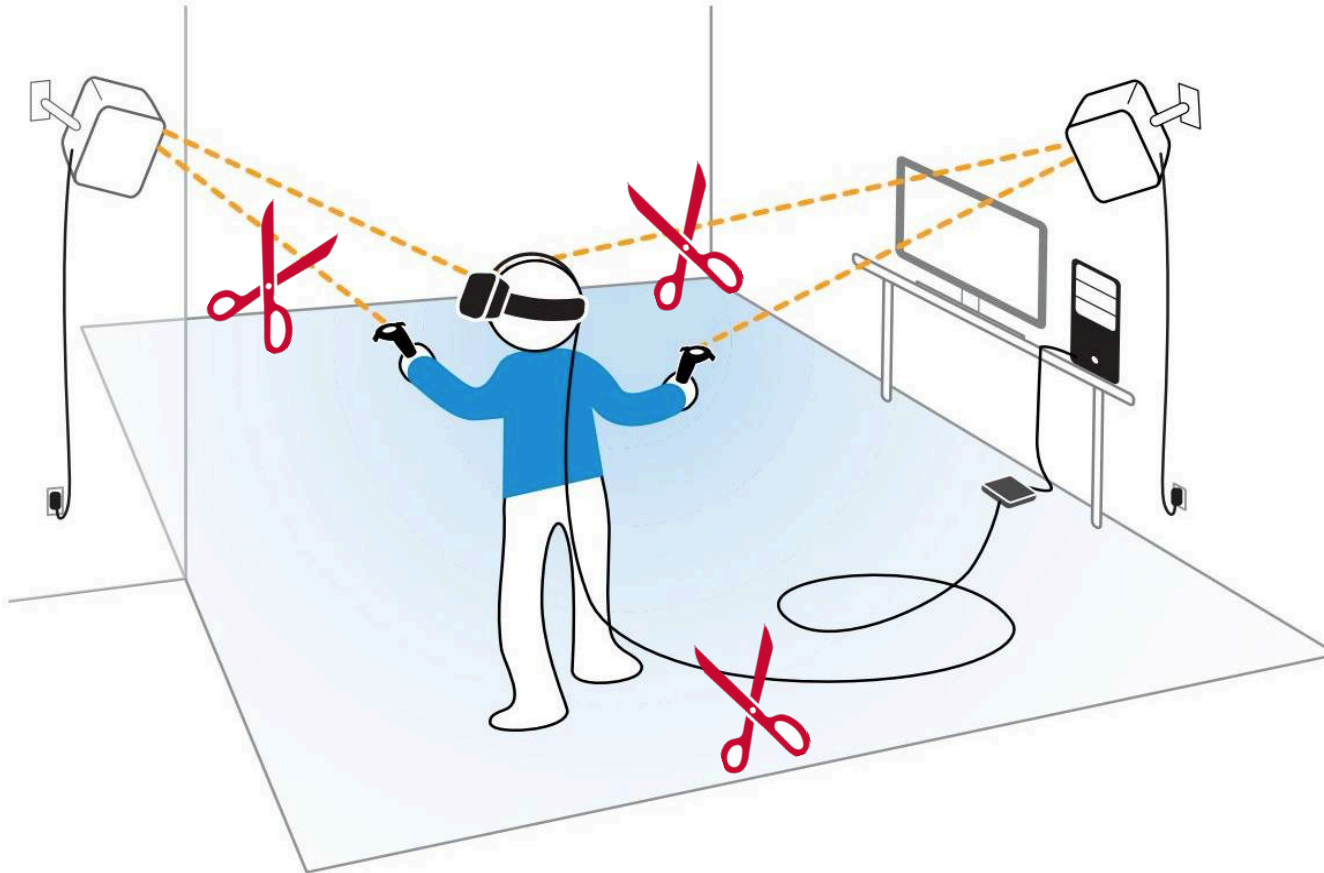


(b) Default Furion Policy

System Architecture

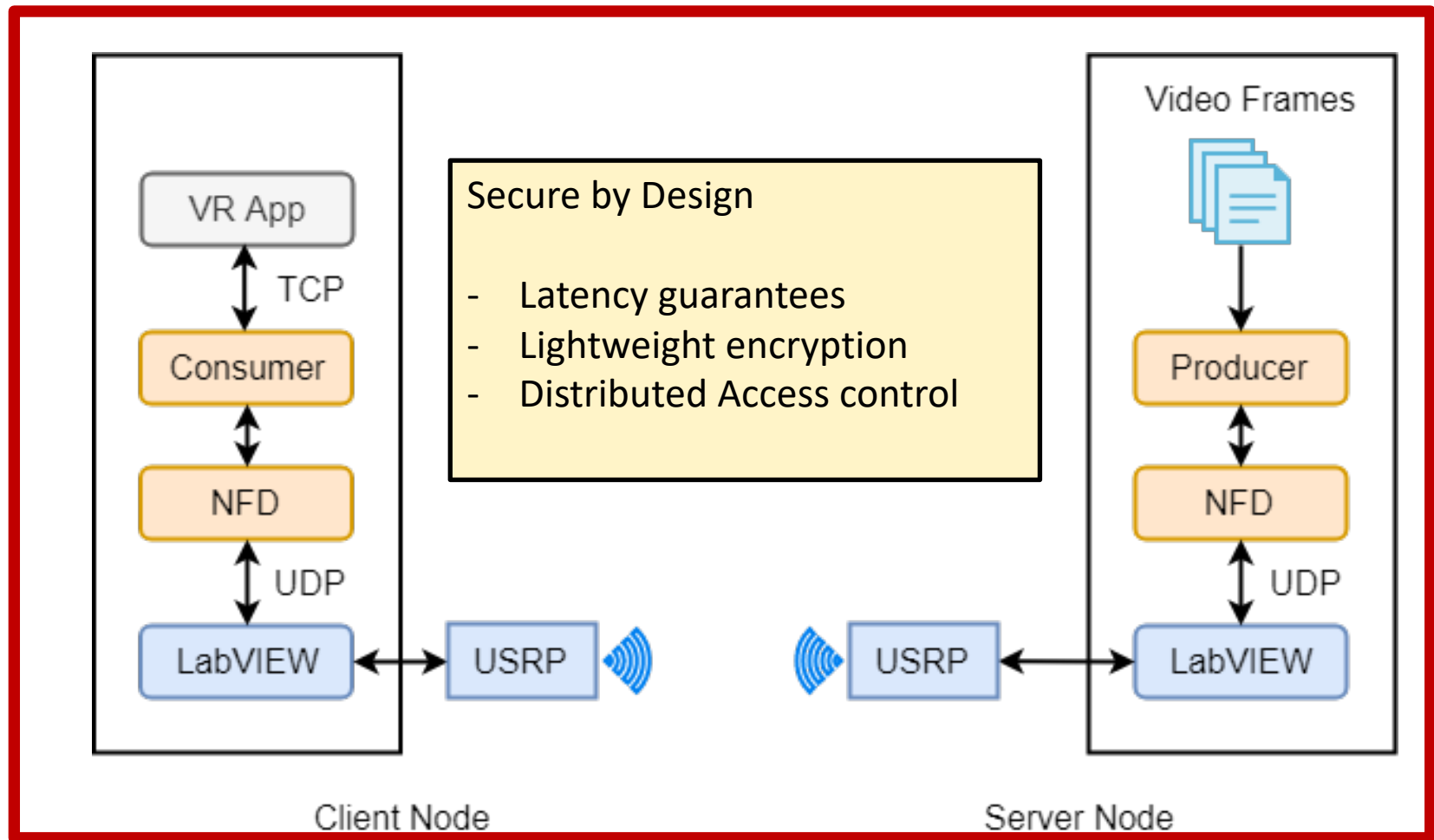


How to cut the cords?



- ❑ Charlie Hu: **High quality video** (enabling high quality VR apps)
- ❑ Romit Roy Choudhury: **Very precise (head) motion** tracking

System Architecture



C. Security

- LTE Inspector – first systematic methodology for the security analysis of 4G protocol
- 5GReasoner – first systematic methodology for the security analysis of 5G protocols
- Torpedo attack – identification of a novel privacy attack exploiting paging occasion in 5G cellular protocol
- PKI approach for digital signature of base station broadcast messages
- Variable paging occasion techniques
- An efficient hierarchical identity-based signature scheme, based on Schnorr signatures for base station broadcast messages
- Aggregate digital signature techniques for NDN

Security of NDN Messages

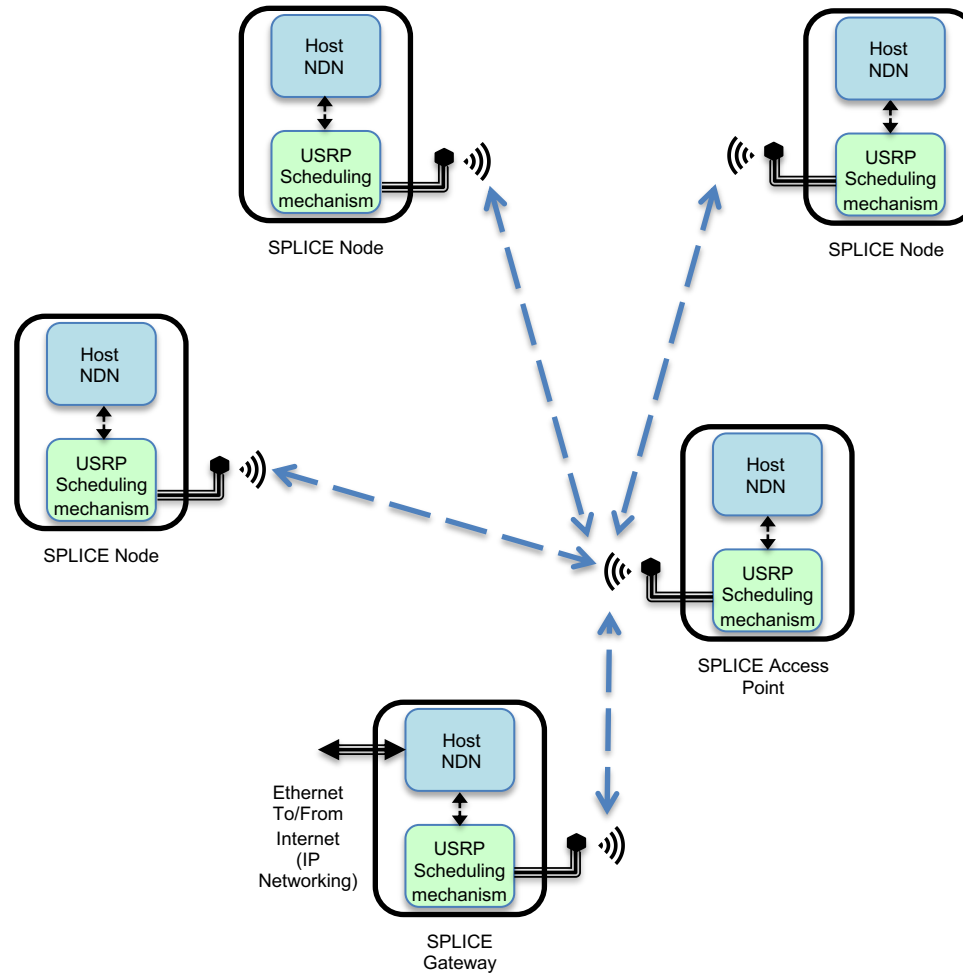
■ Standalone Message Authentication

- 10,000 interest-data packet exchanges between consumer and producer.
- Per 100 signature aggregation.

	ECDSA-256	RSA-3072	SCRA-C-RSA [L=32]	SCRA-C-RSA [L=16]
Public key size	91	422	422	422
Average signature size	71	384	5.82	5.82
Average signing time	0.059	1.49	0.22	0.11
Average verification time	0.10	0.063	0.040	0.02
End-to-end delay	0.35	1.85	0.46	0.30

- Structure-free and Compact Real-time Authentication (SCRA) is able to attain significant latency improvements

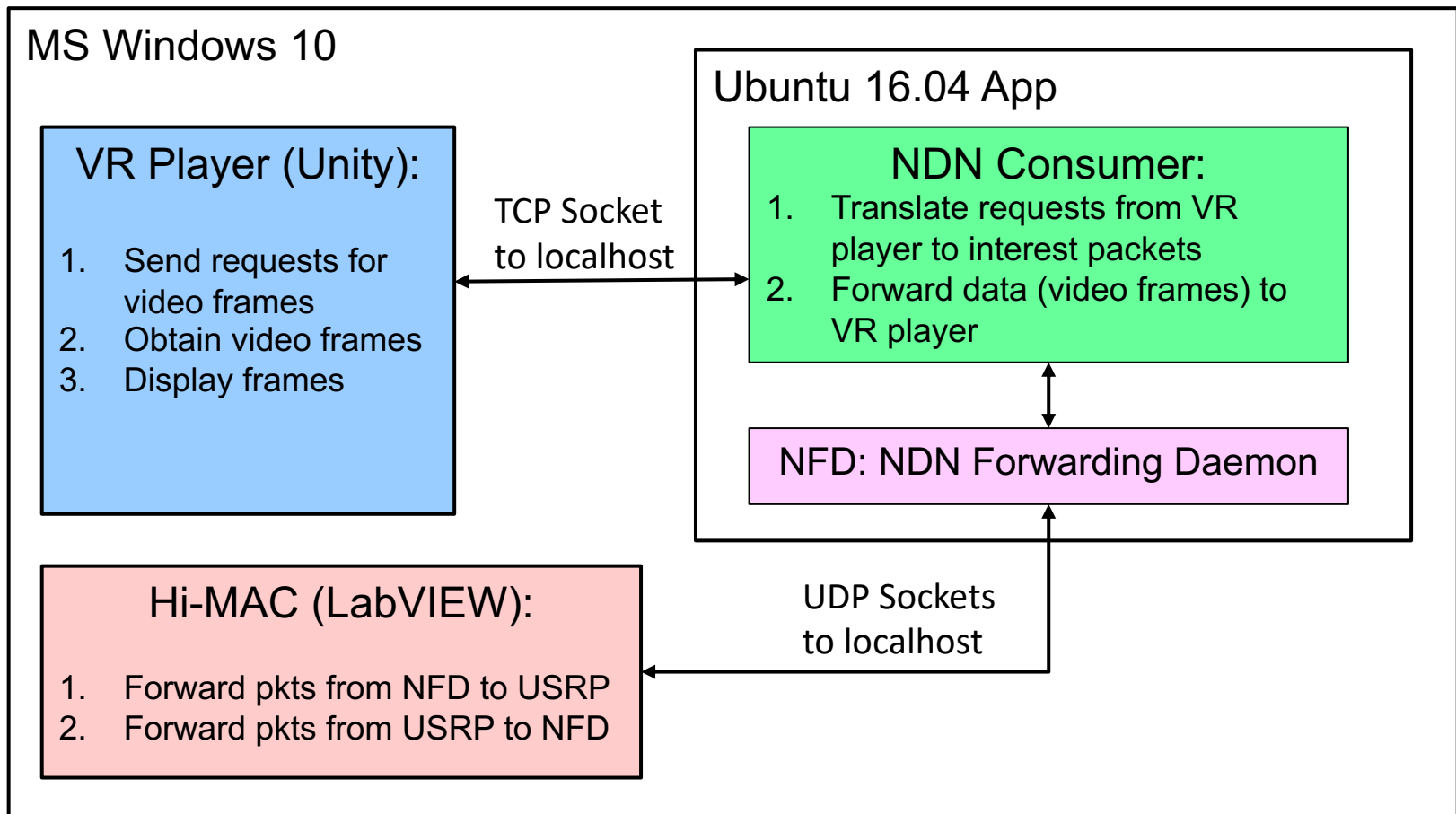
SPLICE Network



Analytically Provable Performance → Measurably Viable Systems

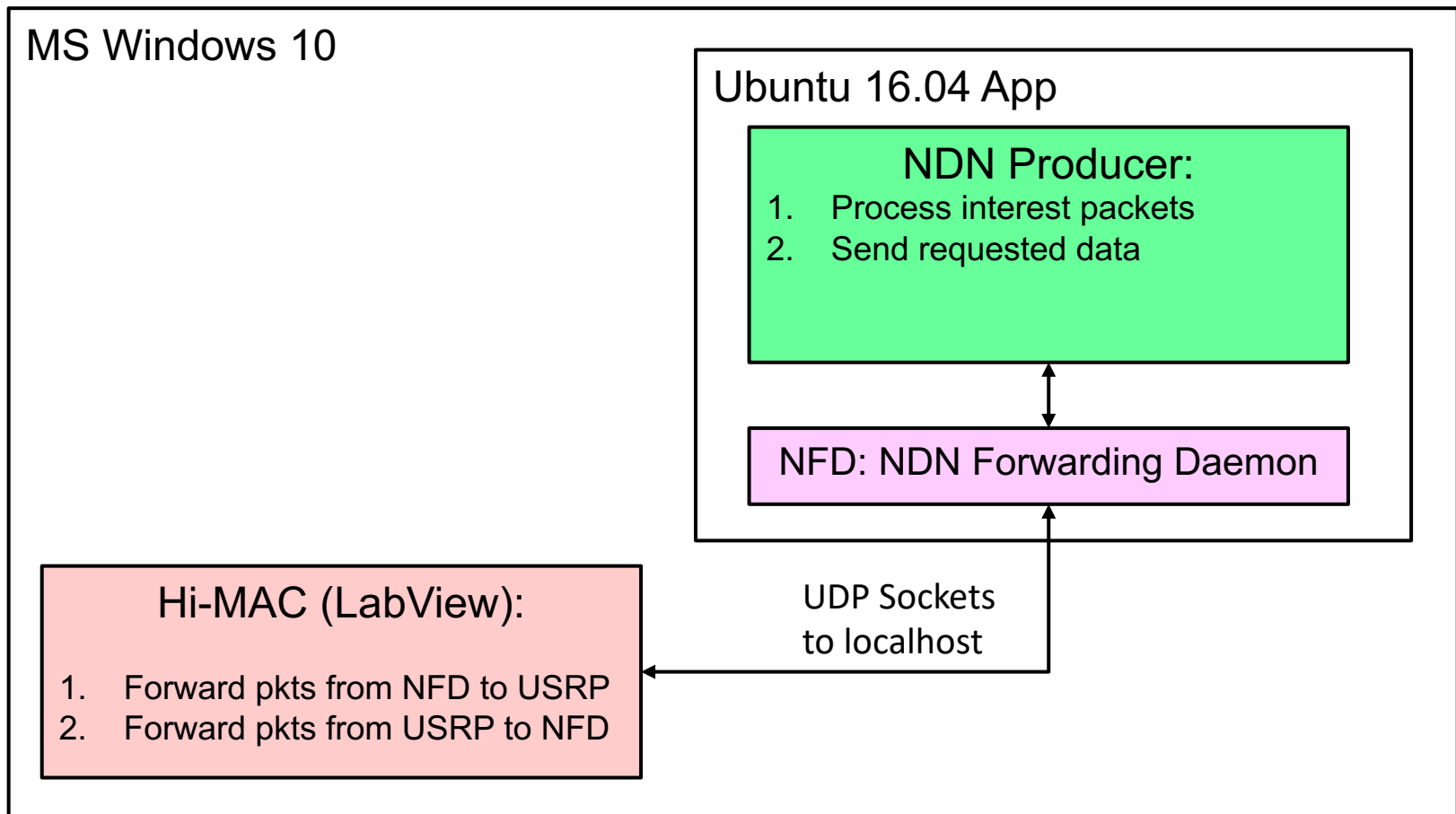
Software Architecture

■ Client side architecture:



Software Architecture

- Server side architecture: (All video frames are generated in advance and stored in the server's hard drive)



Thank you!